

MOTORSPORT AUSTRALIA MANUAL

CIRCUIT RACE APPENDIX

3rd CATEGORY – TOURING CARS

GROUP 3X – SUPERMINI CHALLENGE – TECHNICAL REGULATIONS

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This edition replaces the 2026 appendix. SuperMini Challenge is a single specification. Group 3D Sport Sedan Automobiles based on a BMW Mini may be invited to race with the series under Article 20.

These regulations commence in full on the date in the Modified Articles table. Nothing in this edition requires a change to the cylinder block, crankshaft, conrods, pistons, cylinder head, flywheel, clutch, gearbox, suspension or brakes. The parts that change on an Automobile are the supercharger pulley, the camshaft and the exhaust manifold where they differ from the control items, and the addition of the control restrictor plate and nominated sensors.

Modified Article	Date of Application	Date of Publication
Art. 2, 3, 7.1(a), 7.1(g), 7.3, 7.5(c), (e), (f), (g), 7.8, 8, 9(a), 13(a), 14(a), 15.1, 16.1(b), 17, 18, 19, 20	[commencement date]	[publication date]
Appendix 1, 3, 9, 11, 12, 13, 14	[commencement date]	[publication date]

A capitalised and italicised word in this document is defined in the FIA International Sporting Code or the National Competition Rules, including their Appendices. Any HEADING is for reference only and has no regulatory effect.

Recognised Association: SuperMini Challenge Inc — www.supermini.au

For technical information and control component supply, email: admin@supermini.au

1. PREAMBLE

These regulations apply to each SuperMini Challenge Automobile when entered in a SuperMini Challenge competition.

- (a) Unless authorised by these regulations, each Automobile must remain unmodified and identical to the standard production model as supplied by the manufacturer.
- (b) Any modification not permitted by these regulations is forbidden. A worn or damaged component must be replaced only by a standard production component compliant with these regulations.
- (c) Carbon fibre, carbon/Kevlar composite, ceramic material and titanium alloy are not permitted unless fitted as standard by the manufacturer or expressly permitted. Replacement racing seats are excluded from this regulation.
- (d) Each reference to a Schedule means a Schedule of the Technical Appendix in the Motorsport Australia Manual. Each Automobile must comply with each relevant Schedule.
- (e) Unless specified otherwise, engine machining, finishing and weighing tolerances are per the Motorsport Australia Manual, Technical Appendix – Definitions Technical.
- (f) These regulations should be read with the 3X SuperMini Challenge Technical Guide, which lists control items, part numbers, suppliers and recommended engine and cooling specifications.

DEFINITIONS

Engine bay The area in front of the firewall, behind the radiator support panel, above the bottom of the radiator support panel, below the closed bonnet and between the inner guards.

Commercially available product A product produced in commercial quantities, sold new to be owned outright, stocked by an Australian supplier and available at the same commercial price to any person in Australia for a minimum of one year.

Control item A component, calibration or consumable nominated in Appendix 3. Only the nominated item may be used, in the condition supplied. No modification or substitution is permitted.

Nominated sensor A sensor of the make, part number and mounting position stated in Appendix 3.

Master calibration The SuperMini Challenge ECU calibration file identified by version and checksum in Appendix 13.

Eligibility Officer The person appointed by the SMC committee to administer Articles 3 and 19.

Free The original part and its function may be removed or replaced with a new part, provided the new part has no additional function relative to the original.

AGM battery A sealed lead acid battery of absorbed glass mat construction.

GEL battery A sealed lead acid battery with suspended electrolyte and a silica additive.

2. ELIGIBLE AUTOMOBILES

- (a) A SuperMini Challenge Automobile is a BMW Mini R50 or R53 bodyshell, produced 2001 to 2006, fitted with the W11B16 supercharged engine and driveline as defined in these regulations.
- (b) No other Automobile is eligible.

3. ELIGIBILITY AND INSPECTION

- (a) A scrutineer or the Eligibility Officer may at any time during an Event check the eligibility of an Automobile or Competitor; require an Automobile to be dismantled by the Competitor; require parts, samples or data to be supplied; require a control item to be replaced with one supplied by the Eligibility Officer; or require an Automobile or component to be delivered to a specified location for testing within a specified time after the Event.
- (b) Inspection is a standing right. No protest or reason is required. Components are not sealed.
- (c) The Competitor must demonstrate compliance and must be able to identify the make, model and specification of every regulated component.
- (d) Inspection tiers, penalties and protests are in Article 19.

4. IN CAR VIDEO

- (a) Each Automobile must carry at least one operating forward-facing video camera recording to SD or micro SD, showing the view forward of the Automobile and the driver's steering inputs, mounted as close to horizontal as possible.
- (b) The mount must generally comply with the Motorsport Australia camera technical document. Suction mounts are not permitted.

- (c) On request of the DSO or the management committee, the entrant must provide the video immediately after a race. Failure to do so may be penalised, including rear of grid for the following race.
- (d) A Competitor who does not provide video as required waives all rights to protest against another Competitor and is assumed to be in error.
- (e) SMC retains copyright in all images and video submitted for investigation. Written permission of the SMC committee is required before publication.
- (f) By competing, Competitors grant SMC the irrevocable right to use their name, image, audio and video in any medium for SMC purposes, waive any interest in the copyright, and release SMC from any claim arising from publication.

5. LIVERY

- (a) The SuperMini Challenge windscreen banner and current season sponsor decals are compulsory on every Automobile and may not be replaced.
- (b) Additional sponsor decal positions are advised by the committee at the start of each season.
- (c) One windscreen banner is supplied to each member per season. Replacements may incur a cost. SMC door numbers may be purchased.

6. COMPONENTS

- (a) Any component of any Australian specification Mini Cooper or Mini Cooper S may be used unless otherwise specified.
- (b) Where replacement of a component is permitted, the replacement is free unless restricted by these regulations or unless it is a control item.
- (c) Fasteners, coolant thermostat, indicator and headlight assemblies, belts, gaskets, seals, flexible hoses, coolant pipes, mechanical cables, bearings, clamps, spark plugs and leads, filters, batteries and battery cables, globes and LEDs, fuses, relays and windscreen glass are free, provided no modification is made to fit them and the new part has no additional function.

7. ENGINE

7.1 GENERAL

- (a) The engine must be the W11B16 as fitted to the R53 Mini Cooper S delivered in Australia 2001 to 2006.
- (b) All engine components must be original BMW Mini components, a commercially available product, or the control item, unless otherwise specified.
- (c) Permitted reconditioning: cylinder boring and honing; decking and facing; line boring and honing; conrod resizing; thread repair; replacement of welsh and gallery plugs; cleaning that does not remove original material; crankshaft journal grinding and finishing. Bead blasting and angle machining are not permitted.
- (d) Engine mounts must remain standard in number, location and orientation. Engine mounting bushes are free.
- (e) Air conditioning and heating systems may be removed.
- (f) A crankcase catch can discharging to atmosphere per Schedule B is permitted. Hoses maximum 13 mm inside diameter, connected only to the original breather connections. PCV valve internals may be removed.
- (g) An aftermarket harmonic balancer may be fitted. The crank pulley must be of standard diameter.
- (h) The bypass (diverter) valve must be the standard item or a commercially available replacement of the same function.
- (i) Serpentine belt and tensioner are free provided the system is 6PK and commercially available in Australia.
- (j) A centre main bearing cap strengthening plate per Appendix 7 may be fitted.

7.2 CYLINDER BLOCK, CRANKSHAFT, RODS, PISTONS

- (a) The factory block must be used. Bore may be increased by a maximum of 1.0 mm.
- (b) Pistons are free, forged permitted. Compression ratio 8.5:1 maximum, measured per Appendix 14.
- (c) Conrods are free at standard length.
- (d) The crankshaft must remain standard. Localised spot machining of the counterweights and big end faces for balance only. A maximum of 0.5 mm may be removed from any bearing journal. No other machining, lightening or polishing. Stroke standard.
- (e) Piston rings may be replaced: one compression ring per groove (two) and one three-piece oil ring. Ring contact area not less than standard. Gas-ported rings or pistons are not permitted.

7.3 CYLINDER HEAD, CAMSHAFT, VALVES

- (a) The camshaft must be the control camshaft, unmodified, fitted with the standard camshaft wheel in the standard position. Adjustable, vernier and offset wheels, keys and dowels are not permitted. Lobe lift must be as stated in Appendix 1.
- (b) The cylinder head is free, provided it is a W11B16 casting with the standard camshaft journal positions and the standard number and arrangement of valves. Porting, valves, seats, guides, springs, retainers and gasket faces are free.
- (c) The cylinder head gasket is free.
- (d) Variable valve timing or lift is not permitted.

7.4 LUBRICATION

- (a) An oil cooler is permitted. An aftermarket oil cooling system may replace the factory system.
- (b) Sump baffling is permitted within the removable sump. The oil pickup may be modified.

7.5 INDUCTION

- (a) The air filter system is free upstream of the throttle body. Only ambient air may be supplied. The induction system must be fully contained within the engine bay.
- (b) No liquid, gas or solid other than ambient air may be introduced into the intake, and nothing may be sprayed on or applied to the intercooler. Ice is not permitted.
- (c) The throttle body must be the factory item or Bosch 0 280 750 151. The intake system from the restrictor plate to the cylinder head face must be unmodified.
- (d) The supercharger must be the standard Eaton M45, unmodified. No internal change or flow work.
- (e) The supercharger pulley and the crank pulley must be of standard diameter per Appendix 1, measured at the belt track. No overdrive or underdrive is permitted.
- (f) The intake air temperature sensor must be the nominated sensor in the nominated position per Appendix 3.
- (g) All air entering the engine must pass through the control restrictor plate, Appendix 11, fitted between the throttle body and the supercharger inlet with a gasket each side. The plate must be the item issued to the Automobile, unmodified, with its serial number legible. No air may enter downstream of the plate other than bypass valve recirculation and crankcase breathing through the original connections. The inlet tract between the plate and the supercharger may be pressure or smoke tested at any time.
- (h) An aftermarket intercooler in the factory location is permitted. The bonnet scoop must seal to the intercooler.

7.6 COOLING

- (a) An aftermarket radiator in the factory location is permitted. The plastic radiator support may be replaced with metal on the original mounting points.
- (b) The electric fan must be the factory item. Fan control is free. Coolant hoses, expansion tank, ducting and sealing between the front bumper opening and the radiator, and a radiator screen are free.
- (c) An electric water pump may replace the mechanical pump.
- (d) Coolant per Schedule B.

7.7 FLYWHEEL AND CLUTCH

- (a) Free.

7.8 EXHAUST

- (a) The exhaust manifold must be the control manifold, Appendix 12, unmodified.
- (b) The exhaust system downstream of the manifold flange is free, provided it exits from the rear of the Automobile and the factory fuel tank heat shield is retained.

8. FUEL SYSTEM AND FUEL

- (a) Fuel must be 98 RON pump fuel per Schedule G. No additives. Ethanol content must not exceed that of the fuel as sold.
- (b) A fuel sampling fitting per Appendix 3 must be fitted in the supply line in the engine bay. Fuel may be sampled and tested at any time.
- (c) The original fuel tank must be retained in the original location. Safety foam may be fitted. Tank protection may be added in the area of the tank only.
- (d) Fuel pumps and filters are free, fitted within the tank on the standard hanger or in the standard external location.

- (e) Fuel injectors must be the control injector. The injector harness may be repinned to suit.
- (f) A fuel pressure sensor per Appendix 3 must be fitted to the fuel rail and connected to the ECU.
- (g) The fuel system is otherwise standard.
- (h) The carbon canister may be removed. If removed, the tank must vent externally to the cockpit through a gravity roll-over valve.

9. TRANSMISSION

- (a) The gearbox must be a factory R53 Cooper S Getrag six-speed. Two units were fitted in production: the 2002–2004 (pre-facelift) gearbox and the 2005–2006 (facelift) gearbox, which have different internal ratios and final drives. Either may be used as a complete unit with its original ratios and final drive. Ratios and final drives are listed in Appendix 1. Mixing internal components between the two units, or fitting any other ratio or final drive, is not permitted.
- (b) The gear selector mechanism may be modified to reduce free travel. An aftermarket shifter may be fitted. Factory shift cables must be retained. The shift pattern must not change.
- (c) Limited slip and locked differentials are permitted.
- (d) Pattern aftermarket driveshafts are permitted.

10. CHASSIS

- (a) A safety cage per Schedule J must be fitted.
- (b) Chassis strengthening is permitted only at the locations in Appendix 6.

11. WHEELS AND TYRES

- (a) Wheels are free: one-piece, 17 x 7.5 maximum, all four the same width, diameter and offset.
- (b) The control tyre is the RubberCraft SP-01R 225/45R17.
- (c) Tyre buffing, regrooving and chemical treatment are not permitted.
- (d) Tread depth must exceed 1.5 mm before practice or racing, excluding the shoulder (40 mm from the outer sidewall) and one localised flat spot.
- (e) Wheel spacers per Schedule E, 15.0 mm maximum. Extended studs 90 mm maximum; stud and nut must not protrude past the rim face. Wheel nuts ferrous, open ended.
- (f) The tyre and wheel must be within the perimeter of the Automobile when viewed from above.
- (g) Four new control tyres per round, registered at scrutineering before first practice. Only registered tyres may be used at that round. A registered tyre damaged beyond use may be replaced with the approval of the Eligibility Officer. Tyres are identified by serial and date code. Tyre softeners and any chemical treatment are not permitted and may be tested for.

12. STEERING

- (a) The steering rack, power steering pump and accessories must be standard BMW Mini.
- (b) The steering wheel must not contain wood. A quick-release boss is permitted.

13. BRAKES

- (a) Front calipers and rotors are free, provided the caliper has a maximum of four pistons and the rotor does not exceed 304 mm diameter and 26 mm thickness. Friction material free. Hoses free.
- (b) Rear calipers standard. Rear discs standard size. Rear pads free.
- (c) The factory ABS must be disabled. The actuator may remain.
- (d) The factory master cylinder and booster must be retained.
- (e) The parking brake may be removed. Pad knock-off springs are permitted.
- (f) Brake proportioning valves and brake pressure sensors are not permitted.
- (g) Brake cooling: the fog light or blanking plate in the lower front bumper may be removed. One duct per front wheel, 80 mm maximum inside diameter except within 150 mm of the bumper and 80 mm of the rotor, wholly within the bodywork and not visible from the front. Backing plates may be modified or removed.

14. SUSPENSION

Factory suspension components must be used except as follows.

- (a) Aftermarket adjustable dampers may be used. A control damper package is under consideration by the SMC committee. Should one be adopted, it will be published not less than 24 months before its date of application, and dampers in use at the date of publication remain eligible until that date.
- (b) Each original bush at a suspension pivot point may be replaced only with an elastomeric bush per Definitions – Technical in the Motorsport Australia Manual.
- (c) Front camber plates, fixed or adjustable, are permitted. Fixed metal on the strut tower may be removed to fit them.
- (d) Rear control arms may be replaced with the R56 aluminium item.
- (e) Uprights, hubs, front control arms and rear arms other than toe arms must be OEM. Subframes in the factory location, unmodified except (f).
- (f) Front subframe tapped holes for the rear castor bush mounts and front inner ball joint may be drilled to accept through bolts and nuts, grade 8.8 minimum, for serviceability only.
- (g) Maximum track 1750 mm.
- (h) Aftermarket anti-roll bars are permitted on the original mounting points on both the chassis and the control arm. Adjustable anti-roll bars are not permitted. One front and one rear link may be disconnected.
- (i) A front strut brace linking only the strut towers is permitted. Rear towers may be braced by the cage and/or a strut brace.

15. ELECTRICAL

15.1 ENGINE MANAGEMENT

- (a) Control ECU: Link G4X Plugin, SKU 209-4000, in the factory ECU location, on the firmware version in Appendix 13. CAN transmit only.
- (b) The master calibration must be installed and unaltered. It is loaded and locked by the Eligibility Officer. Competitor-adjustable parameters are limited to: idle speed, cold start enrichment, fan and pump control, and gauge and dash outputs. All other parameters are locked.
- (c) The calibration checksum and firmware version may be read at any time and must match Appendix 13. A mismatch is non-compliance under Article 19.
- (d) Nominated sensors per Appendix 3 must be fitted in the nominated positions: manifold pressure, intake air temperature, coolant temperature, wideband lambda, fuel pressure. No other sensor may influence fuel or ignition.
- (e) ECU internal logging must be enabled per the master calibration and must be continuous from first practice to the end of the last race. A log that has been cleared, interrupted or restarted during an Event is non-compliance unless the Eligibility Officer is satisfied of the cause. The log may be downloaded by the Eligibility Officer at any time.
- (f) Rev limit 7,000 rpm, set in the master calibration.
- (g) Traction control, launch control, anti-lag and torque management functions are not permitted.
- (h) Ignition coils, igniters and triggers to OE specification. Spark plugs and leads free.
- (i) The master calibration is the property of SMC. It is issued to Competitors as a condition of entry and may not be copied, modified or supplied to any other person. Updates are issued to all Competitors together on a published date.

15.2 GENERAL

- (a) A rain light must be fitted centrally on the rear hatch above the number plate recess.
- (b) The battery may remain in the factory location or be relocated to the cockpit floor. AGM or GEL only. Isolator required. Securely mounted, positive terminal insulated.
- (c) The factory harness may be modified or replaced.

15.3 DATA AND CAMERAS

- (a) Data logging: standalone beacon lap timer, standalone GPS lap timer, or a solidly mounted phone lap timer only. No connection to the ECU, OBD port or vehicle wiring other than power, ground and RPM.
- (b) One video system on a solid mount to the safety cage, commercially available.
- (c) The timing transponder must be mounted under the left front footwell.

16. BODY, COACHWORK AND COCKPIT

16.1 EXTERIOR

- (a) All coachwork standard. A rear wing may be fitted: standard R53, TDI01, RSIC6, or identical.
- (b) Wheel arch lips may be folded or trimmed for tyre clearance. Panel shape, profile and width must remain. Plastic inner guard liners may be removed. Wheel arch trims must be the standard item or a replacement of standard shape, profile and width; widened or flared arches are not permitted.
- (c) Plastic undertrays may be removed. No additional underbody panels.
- (d) Front and rear bumpers unmodified except for tow points. Metal reinforcements retained. No elements added below or behind the covers. Lightweight stays for bumper corners and an indicator support are permitted.
- (e) The rear wiper motor must be removed and the hole covered.
- (f) Bonnet sound deadening, side mouldings, rear fog light and bonnet sealing rubber may be removed. Resulting holes must be covered.
- (g) Original bonnet latches must be removed. Bonnet vents are permitted. Ducting from the bonnet to the air filter is not permitted. The factory scoop may be carbon fibre.
- (h) Only the original rear tow point, with optional strap or shackle.
- (i) Fastener holes of minimum size may be drilled. No additional ventilation holes to the front bar area.
- (j) No fixed metal may be removed except for cage or suspension installation. Unused brackets may be removed.

16.2 WINDOWS

- (a) Original glazing retained. Sunroof may be removed and the hole covered with aluminium, steel or FRP per the Manual. Both wipers must operate.

16.3 INTERIOR

- (a) May be removed: interior trim and glove box, roof lining, carpet and insulation, passenger and rear seats, radio and console, restraint and airbag systems, boot lining, parcel shelf, spare wheel and tools. Airbags must be removed.
- (b) Heating and air conditioning components may be removed or modified.
- (c) The factory pedal assembly must remain. Pedal settings may be adjusted on the original mounts.
- (d) Door and quarter trims may be replaced with solid material.
- (e) The dashboard remains in the factory location, modified only for cage clearance.
- (f) Permitted additions: safety equipment, switches and dials, driver cooling, drink system, ballast per Article 17, two-way radio (no car-to-car), electrical equipment per Article 15, a footwell floorplate, and a knee guard per Appendix 8.
- (g) Nothing may hinder driver vision or extrication or affect the engine, steering, transmission, brakes or roadholding. All items must be secured.
- (h) Each control retains its standard function. Seat tracks may be reinforced or removed to fit a racing seat.

17. WEIGHT

- (a) Minimum racing weight 1,200 kg with driver, as presented at any time during an Event.
- (b) Ballast must be in the cockpit, visible, accessible and mounted per the Manual requirements for ballast retention.

18. MISCELLANEOUS

- (a) Competition numbers are issued by the SMC registrar.
- (b) Each Automobile must be registered with the SMC registrar, with its ECU serial number, restrictor plate serial number and control item details, before it competes.

19. INSPECTION, PROTEST AND PENALTIES

- (a) Tier 1 – pit lane. Weight, track, tyres, restrictor plate serial and bore, inlet tract integrity, pulley diameters at the belt track, exhaust manifold, throttle body, injector, sensors and wiring between sensors and ECU, ECU serial, calibration checksum and firmware, log download and continuity, fuel sample, camera, brake rotor size. Random draw and the round winner at each round, before or after any session.
- (b) Tier 2 – cam cover off. Camshaft identification, lobe lift and timing. A cranking compression test may be taken as a screening measure and may be used to direct an Automobile to Tier 3. Any Automobile in the top five of the

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championship, after the last race of a meeting, at the Eligibility Officer's discretion. Cam cover gasket supplied by SMC.

- (c) Tier 3 – head off. Compression ratio per Appendix 14, bore, head casting and valve count. The championship winner's Automobile, within 14 days of the final round, at a nominated workshop, funded by SMC. The championship is not declared until cleared. Any other Automobile may be inspected at Tier 3 on protest or at the Eligibility Officer's direction.
- (d) ECU logs may be collected from any Automobile at any time and compared across the field. A log showing operation outside the master calibration is non-compliance.
- (e) The ECU log and in-car video of the pole-position Automobile for each Event are published to all Competitors after the Event.
- (f) Non-compliance forfeits all points from the meeting, or from the season if found at the season inspection. Refusal of inspection is non-compliance. A false statement to an Eligibility Officer or scrutineer is a conduct matter.
- (g) Protests per the NCR. Fees per Appendix 1. On an upheld protest the fee is returned and the protested Competitor bears the inspection cost. On a dismissed protest the fee passes to the protested Competitor and the protester bears the inspection cost. Club inspection is free to a compliant Competitor.
- (h) Non-compliance is determined by the Stewards on the Eligibility Officer's recommendation. The Motorsport Australia appeal path applies.
- (i) No parity, handicap, success ballast or performance adjustment applies.

20. INVITED AUTOMOBILES

- (a) Group 3D Sport Sedan Automobiles based on a BMW Mini may be invited to compete in SuperMini Challenge races as stated in the Supplementary Regulations for an Event. They comply with Group 3D regulations, are classified separately and do not score SuperMini Challenge points.
- (b) Where the SuperMini Challenge field consistently exceeds fifteen Automobiles, SMC may request a separate grid for invited Automobiles within the same Event.

APPENDIX 1 – DIMENSIONS, LIMITS AND FEES

Figures marked [] are to be set from measurement of standard W11B16 components and the dyno session before publication.

Item	Figure	Checked at
Cylinder bore, standard	77.0 mm	Reference
Cylinder bore, maximum	78.0 mm	Tier 3
Stroke	85.8 mm, standard	Tier 3
Compression ratio, maximum	8.5:1	Tier 3
Control camshaft lobe lift, intake / exhaust	[] / [] mm	Tier 2, dial gauge
Supercharger pulley diameter at belt track	[] mm, standard	Tier 1
Crank pulley diameter at belt track	[] mm, standard	Tier 1
Control restrictor, orifice diameter	[] mm, per Appendix 11	Tier 1, plug gauge
Gearbox, 2002–2004 unit: ratios 1–6 / final drive	[] / [], per BMW data	Tier 1, ratio check
Gearbox, 2005–2006 unit: ratios 1–6 / final drive	[] / [], per BMW data	Tier 1, ratio check
Rev limit	7,000 rpm	Tier 1, calibration
Wheels, maximum	17 x 7.5	Tier 1
Track, maximum	1750 mm	Tier 1
Front brake rotor, maximum	304 mm diameter, 26 mm thickness	Tier 1
Front brake caliper, maximum	4 pistons	Tier 1
Minimum racing weight with driver	1,200 kg	Tier 1
Tyre allocation per round	4 new	Tier 1
Protest fee, Tier 1	[\$300]	
Protest fee, Tier 2	[\$750]	
Protest fee, Tier 3	[\$2,500]	

APPENDIX 2 – RESERVED

Not used.

APPENDIX 3 – CONTROL ITEMS AND NOMINATED SENSORS

Part numbers, suppliers and current prices are published in the 3X SuperMini Challenge Technical Guide.

Item	Article	Nominated item	Effective
Camshaft	7.3(a)	[to be nominated]	[commencement]
Inlet restrictor plate	7.5(g)	SMC plate per Appendix 11, serialised	[commencement]
Intake air temperature sensor and position	7.5(f), 15.1(d)	[part number], [position, e.g. intercooler outlet plenum]	[commencement]
Exhaust manifold	7.8(a)	[to be nominated]	[commencement]
Fuel sampling fitting	8(b)	[type and coupling]	[commencement]
Fuel injector	8(e)	[to be nominated]	[commencement]
Fuel pressure sensor	8(f), 15.1(d)	[part number]	[commencement]
Manifold pressure sensor	15.1(d)	[part number], standard position	[commencement]
Coolant temperature sensor	15.1(d)	Standard	Current
Wideband lambda sensor and controller	15.1(d)	[part number]	[commencement]
Tyre	11(b)	RubberCraft SP-01R 225/45R17	Current
ECU	15.1(a)	Link G4X Plugin 209-4000	Current

Item	Article	Nominated item	Effective
Master calibration	15.1(b)	Per Appendix 13	[commencement]
Damper package	14(a)	Under consideration – not nominated	—

APPENDIX 4 – BRAKE CIRCUIT

Photographs as per the 2026 appendix.

APPENDIX 5 – DELETED

Component sealing is not used. The approved sealers list is withdrawn.

APPENDIX 6 – CHASSIS STRENGTHENING

Diagram as per the 2026 appendix.

APPENDIX 7 – ENGINE BEARING CAP

Centre main bearing cap strengthening plate. Diagram as per the 2026 appendix.

APPENDIX 8 – KNEE GUARD

Maximum 220 mm high and 600 mm long. Diagram as per the 2026 appendix.

APPENDIX 9 – DELETED

Valve throat machining limits no longer apply. The cylinder head is free under Article 7.3(b).

APPENDIX 10 – REAR BUMPER LOWER MOUNTING POINT

Diagram as per the 2026 appendix.

APPENDIX 11 – CONTROL RESTRICTOR PLATE

- Aluminium plate, [8–10] mm thick, cut to the throttle body flange outline and bolt pattern, with a single central orifice of [] mm diameter. Entry radius [] mm on the throttle body side; sharp edge on the supercharger side. Orifice cylindrical for a minimum length of 3 mm.
- Plates are manufactured in one batch to the SMC drawing and issued by the Eligibility Officer. Each plate is stamped with its serial number and orifice diameter.
- Fitted between the throttle body and the supercharger inlet elbow with a gasket each side, using the throttle body fasteners. Where the throttle body flange differs from the elbow, the plate also serves as the adaptor and no other adaptor is permitted.
- The orifice is checked with a plug gauge with the throttle body removed. A plate that admits a gauge larger than the nominated diameter, or that shows any modification, is non-compliance.
- A damaged plate is returned to the Eligibility Officer for replacement.

[Drawing: face view and section through the joint – see restrictor-plate.svg. Photograph: plate installed between throttle body and elbow.]

APPENDIX 12 – CONTROL EXHAUST MANIFOLD

The control manifold is [make, part number]. It must be fitted unmodified, including the collector and flange. Coatings and heat wrap are permitted. Damage may be repaired by welding without change to the tube length, diameter or route.

[Photograph: control manifold with identifying feature marked.]

APPENDIX 13 – MASTER CALIBRATION

Item	Value
ECU firmware version	[]

Item	Value
Calibration file name and version	SMC-2027-[]
Checksum	[]
Issued	[date]
Competitor-adjustable parameters	Idle speed; cold start enrichment; fan control; fuel pump control; gauge and dash outputs
Logging	Enabled, [rate] Hz, channels per the file. Not to be cleared during an Event
Strategy	Closed-loop lambda with long-term trim; knock control active; ignition corrected for intake air and coolant temperature; rev limit 7,000 rpm

[Screenshot: PCLink checksum and firmware screen, for scrutineer reference.]

APPENDIX 14 – INSPECTION PROCEDURES

A Tier 1 – pit lane

Weight on SMC scales with driver. Track. Tyre registration. Restrictor plate serial by inspection; orifice by plug gauge with throttle body removed if directed. Inlet tract between plate and supercharger by inspection and, if directed, pressure or smoke test. Supercharger and crank pulley diameter by vernier at the belt track. Exhaust manifold and throttle body by inspection against Appendix 3 and 12. Injector and sensor part numbers by inspection. ECU serial, firmware and calibration checksum by laptop; log download and check of continuity. Harness between sensors and ECU by inspection. Fuel sample from the sampling fitting; a proportion of samples each season are laboratory tested. Camera operating. Front rotor diameter and thickness by vernier.

B Tier 2 – cam cover off

Cam cover removed by the Competitor in the presence of the Eligibility Officer. Camshaft identified against the control item by its markings; lobe lift by dial gauge against Appendix 1 if directed. Timing checked at TDC against the standard wheel position. Cranking compression may be recorded on all cylinders as a screen. Cover refitted with the SMC-supplied gasket.

C Tier 3 – head off

Cylinder head removed at the nominated workshop. Compression ratio determined by measuring the combustion chamber volume with the head on the bench, the piston crown volume and deck height in the block, and the compressed gasket thickness, calculated per the Motorsport Australia Manual. Bore measured. Head casting number and valve count recorded. Head refitted at SMC cost with a new gasket.

[Photographs required for the Technical Guide: control camshaft markings; standard pulleys with vernier; control manifold; IAT sensor position; fuel sampling fitting; injector and lambda sensor labels; ECU checksum screen.]